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EXPERIMENT-3- Implement a CSS Grid Based layout to create a responsive web page.

AIM:

To Implement a CSS Grid Based layout to create a responsive web page.

PROCEDURE:

1. Create a basic HTML document and add a container for the layout.
2. Link an internal CSS file to the HTML document using style tags.
3. Link Bootstrap and Javascript via CDN.
4. Set the container's display property in CSS.
5. Use display properties / properties to specify the layout arrangement.
6. Assign each child element to its respective area using the property.
7. Apply styling such as background colors, padding, and borders to distinguish sections.
8. Test the webpage and screen resolutions to ensure responsiveness.

PROGRAM CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Responsive Layout</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body {
      background-color: #f4f6f8;
    }
  </style>
</head>
<body>
```

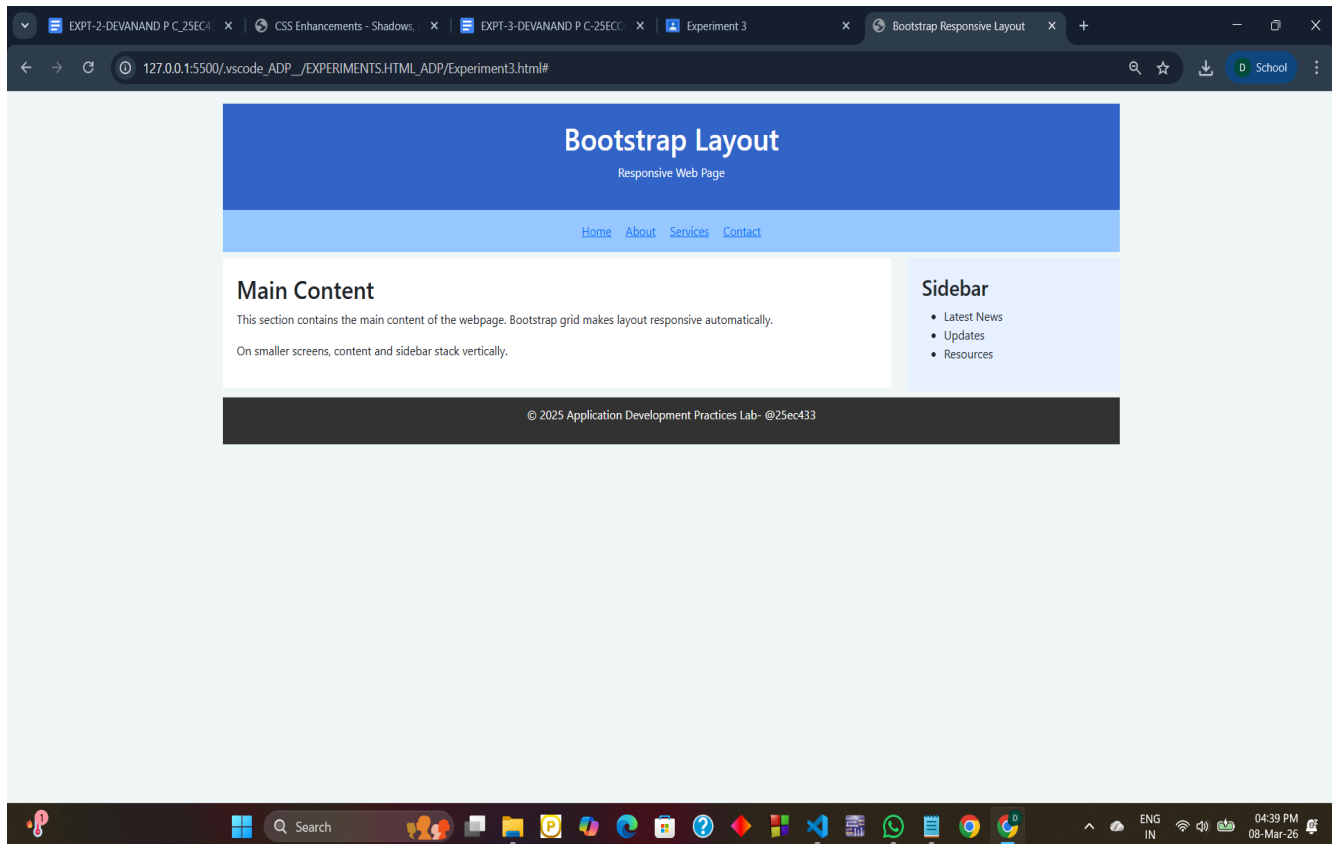
```
header {
  background-color: #3366cc;
  color: white;
  padding: 20px;
  text-align: center;
}
nav {
  background-color: #99ccff;
  padding: 15px;
  text-align: center;
}
main {
  background-color: white;
  padding: 20px;
}
aside {
  background-color: #e6f0ff;
  padding: 20px;
}
footer {
  background-color: #333;
  color: white;
  text-align: center;
  padding: 10px;
}
</style>
</head>

<body>
<div class="container my-3">
  <div class="row">
    <div class="col-12">
      <header>
        <h1>Bootstrap Layout</h1>
        <p>Responsive Web Page</p>
      </header>
```

```
</div>
</div>
<div class="row">
  <div class="col-12">
    <nav>
      <a href="#" class="mx-2">Home</a>
      <a href="#" class="mx-2">About</a>
      <a href="#" class="mx-2">Services</a>
      <a href="#" class="mx-2">Contact</a>
    </nav>
  </div>
</div>
<div class="row mt-2">
  <div class="col-12 col-md-9">
    <main>
      <h2>Main Content</h2>
      <p>
        This section contains the main content of the webpage.
        Bootstrap grid makes layout responsive automatically.
      </p>
      <p>
        On smaller screens, content and sidebar stack vertically.
      </p>
    </main>
  </div>
  <div class="col-12 col-md-3">
    <aside>
      <h3>Sidebar</h3>
      <ul>
        <li>Latest News</li>
        <li>Updates</li>
        <li>Resources</li>
      </ul>
    </aside>
  </div>
</div>
```

```
<div class="row mt-2">
  <div class="col-12">
    <footer>
      <p>&copy; 2025 Application Development Practices Lab- @25ec433</p>
    </footer>
  </div>
</div>
</div>
</body>
</html>
```

OUTPUT:



RESULT:

A responsive web page layout was successfully implemented using CSS Grid and bootstrap.